Note Title 8/19/2025

COL774 / COL7341
Machine Learning
Aug 19, 2025

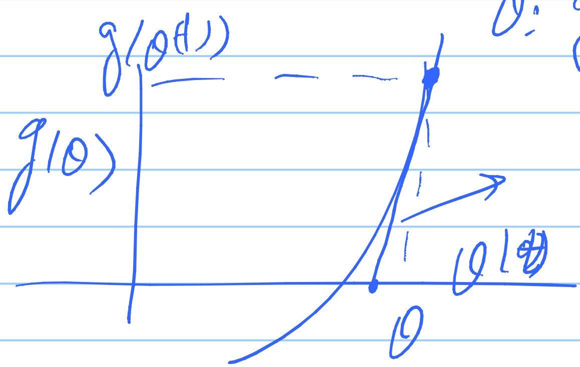
$g: \mathbb{R} \to \mathbb{R}$

Last Class: SGD.

Newton's Method:
(for optimization)
(B)

Newton's method (for finding zeros of a fn.) — (A)

$\theta : g(\theta) = 0$




$$\frac{\underbrace{g(\theta^{(t+1)})}_{=0} - g(\theta^{(t)})}{\theta^{(t+1)} - \theta^{(t)}} = \nabla_\theta g(\theta)\Big|_{\theta^{(t)}} \qquad \frac{\partial g(\theta)}{\partial \theta}\Big|_{\theta^{(t)}}$$

$\theta^{(t)}, \theta^{(t+1)} \ldots$

$$\Rightarrow \quad \underbrace{\theta^{(t+1)}} = \theta^{(t)} - \left[\nabla_\theta\, g(\theta^{(t)})\right]^{-1} \underbrace{g(\theta^{(t)})} \quad \text{— (1)}$$

Newton's update (A)

Objective – Minimize $f(\theta)$

$f: \mathbb{R} \to \mathbb{R}$

Looking for zeros of $f'(\theta)$ $\quad \dfrac{\partial f(\theta)}{\partial \theta}$

$$\theta^{(t+1)} \leftarrow \theta^{(t)} - \left[\frac{\partial^2 f(\theta)}{\partial \theta^2}\right]^{-1} f'(\theta^{(t)}) \quad ✓ \text{ (2)}$$

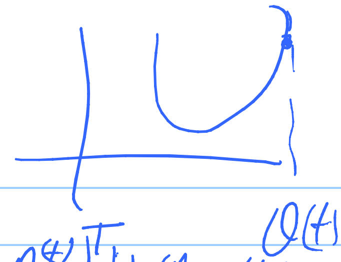
$\to \left.\dfrac{\partial f(\theta)}{\partial \theta}\right|_{\theta^{(t)}}$

In higher dimensions:

$f: \mathbb{R}^n \to \mathbb{R}$

$$\theta^{(t+1)} \leftarrow \theta^{(t)} - H^{-1} \left.\nabla_\theta f(\theta)\right|_{\theta^{(t)}} \to \text{Newton's update} \text{ — (3)}$$

$f : \mathbb{R}^n \to \mathbb{R}$




$$\hat{f}(\theta) = f(\theta^{(t)}) + (\theta - \theta^{(t)})^T \nabla_\theta f(\theta^{(t)}) + \frac{1}{2} (\theta - \theta^{(t)})^T H (\theta - \theta^{(t)})$$

$\Downarrow$ Quadratic approximation to $f(\theta)$ at $\theta^{(t)}$

$\theta$ : s.t. $\nabla_\theta \hat{f}(\theta) = 0$

$\theta \in \mathbb{R}^n$

$A \in \mathbb{R}^{n \times n} \to$ symmetric

$\nabla_\theta \, \theta^T A \theta = 2A\theta$ (HW)

$\frac{\partial \, \theta^T A \theta}{\partial \theta_j}$ $\forall j$

Differentiating RHS, we get $\nabla_\theta \hat{f}(\theta)$ $\Downarrow$

$$0 + \nabla_\theta f(\theta^{(t)}) + \frac{1}{\cancel{2}} \times \cancel{2} \, H (\theta - \theta^{(t)})$$

$\Rightarrow$ Equating this to zero:

$$H(\theta - \theta^{(t)}) = \nabla_\theta f(\theta^{(t)})$$

$\nabla^2_\theta \underbrace{\theta^T A \theta} = 2A$ ✓

$\rightarrow \theta - \theta^{(t)} = -H^{-1} \nabla_\theta f(\theta^{(t)})$

$$\boxed{\theta = \theta^{(t)} - H^{-1} \nabla_\theta f(\theta^{(t)})}$$

↳ $\theta^{(t+1)}$


$f(\theta) = (\theta - 3)^2 + 1$

$\theta = 4$

$4 - \frac{1}{\cancel{2}} \cancel{2} = \underline{3}$

$\Rightarrow$ **Normalization of data:** $\{x^{(i)}, y^{(i)}\}_{i=1}^{n}$ $x^{(i)} \in \mathbb{R}^n$ $y^{(i)} \in \mathbb{R}$

↳ Normalize data to 0 mean & unit variance $\underline{\forall j}$

$\forall j$: mean of $\mu_j \leftarrow \frac{1}{m}\sum x^{(i)}_j$ ($j^{th}$ mean)

variance $\sigma_j^2 \leftarrow \frac{1}{m}\sum_{i=1}^{m}(x^{(i)}_j - \mu_j)^2$

$\forall i$: — $\forall j$, $\left[ x^{(i)}_j \leftarrow \frac{(x^{(i)}_j - \mu_j)}{\sigma_j} \right]$

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$\Rightarrow$ Closed form solution for Linear Regression

$$J(\theta) = \frac{1}{2m}\sum_{i=1}^{m}\left[y^{(i)} - \theta^T x^{(i)}\right]^2$$